P. R. POTE PATIL

College of Engineering & Management, Amravati.

(An Autonomous Institute)

Department of Artificial Intelligence & Data Science

Year: 2025-26 **Semester: IV**

Course: Computer Network Lab (AI408PCC11**)**

Institute Vision Mission

Vision

To flourish as a center of excellence for producing the skilled technocrats and committed human beings.

Mission

- To create conducive environment for teaching &learning.
- To impart quality education t
- hrough demanding academic programs.
- To enhance career opportunities by exposure to Industries & recent technologies.
- To develop professionals with strong ethics and human values for the betterment of society.

Department of Artificial Intelligence & Data Science Vision

To flourish as a center of excellence for producing the skilled of Artificial Intelligence and committed human beings.

Mission

- To impart quality electrical engineering education through demanding academic and skilled based programs
- To enhance career opportunities in Artificial Intelligence by exposure to Industries & recent technologies
- To develop professionals with strong ethics and human values for the betterment of Society.

Program Outcomes:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8)

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i)independent and life-long learning ii) adaptability to new and emerging technologies and iii)critical thinking in the broadest context of technological change. (WK8)

Program Education Objectives

To prepare the Engineering graduates to,

PEO1: Technical Proficiency: Graduates will demonstrate proficiency in core areas of computer science, including algorithms, data structures, software engineering, and database systems, with a specialized focus on artificial intelligence and machine learning techniques.

PEO2: Leadership and Innovation: Graduates will exhibit leadership qualities and innovative thinking, contributing to the development and implementation of AI-driven solutions that address societal, industrial, and environmental challenges.

PEO3: Ethical and Professional Responsibility: Graduates will adhere to ethical and professional standards in their engineering practice, demonstrating integrity, accountability, and a commitment to societal well-being in the design and deployment of AI and DS systems.

PEO4: Lifelong Learning: Graduates will adopt a culture of lifelong learning and professional development, empowering individuals to adapt to emerging technologies, navigate complex socio-technical landscapes, and contribute meaningfully to the advancement of AI&DS throughout their careers.

Program Specific Outcomes

To prepare the Engineering graduates to,

PSO1: Intelligent Systems Development: Develop intelligent systems and applications integrating AI and DS technologies to enhance functionality and performance.

PSO2: Problem-Solving Skills: Possess strong analytical and problem-solving skills, enabling them to identify, formulate, and solve complex engineering problems using AI and DS methodologies in diverse application domains.

INSTRUCTIONS FOR STUDENTS

1. Students shall read the points given below for understanding the theoretical concepts and practical applications.
2. Listen carefully to the lecture given by teacher about importance of subject, curriculum philosophy, learning structure, skills to be developed, information about equipment, instruments, procedure, method of continuous assessment, tentative plan of work in laboratory and total amount of works to be done in a semester.
3. Student shall undergo study visit of the laboratory for types of equipment, and material to be used, before performing experiments.
4. Read the write up of each experiment to be performed, a day in advance.
5. Organize the work in the group and make a record of all observations.
6. Understand the purpose of experiment and its practical implications.
7. Student should not hesitate to ask any difficulty faced during conduct of practical /exercise.
8. Write the answers of the questions allotted by the teacher during practical hours if possible or afterwards, but immediately.
9. Student should develop the habit of pear discussion / group discussion related to experiments / exercise so that exchanges of knowledge / skills could take place.
10. Students shall attempt to develop related hands-on-skills and gain confidence.
11. Student shall focus on development of skills rather than theoretical or codified knowledge.
12. Student shall insist for the completions of recommended Laboratory Work, answers to the given question etc.
13. Student shall develop the habit of evolving more ideas, innovations, skills etc. that included in the scope of the manual.
14. Student shall refer technical magazines, proceedings of the Seminars, refer website related to the scope of the subjects and update their knowledge and skills.
15. Student should develop the habit of not depend totally on teachers but to develop self-learning techniques.
16. Student should develop the habit to interact with the teacher without hesitation with respect to academic involved.
17. Student should develop habit to submit the practical's exercise continuously and progressively on the schedule dates and should get the assessment done.
18. Student should be well prepared while submitting the write up of the exercise. This will develop the continuity of the studies and he will not be overloaded at the end.

P. R. POTE PATIL

COLLEGE OF ENGINEERING & MANAGEMENT, AMRAVATI.

Department of Artificial Intelligence & Data Science

Certificate

This is to certify that Mr./ Ms…… of …….. Semester of Bachelor of Engineering in Artificial Intelligence & Data Science of P. R. Pote Patil College of Engineering & Management, Amravati, has completed the term work satisfactory of course …….. for the academic year 20 - 20 as prescribed in the curriculum.

Date…… **Roll No……**

Subject Teacher **Head of the Department**

LIST OF (PRACTICALS / EXPERIMENTS) & PROGRESSIVE ASSESSMENT FOR TERM WORK

Academic Year :2025-26 **Course: Computer Network Lab**

Course Code: (AI408PCC11)

Semester: IV

Name of Faculty: ……

Name of Student: ……

Roll No ……

SN.	Title of the Practical / Experiment	Date of Performance	Date of Submission	Sign of Teacher
1	Aim: To study and identify the physical topology and hardware components of a LAN			
2	Aim: To configure IP addresses for network nodes and verify connectivity.			
3	Aim: To execute common network diagnostic commands and interpret their results			
4	Aim: To install and configure OMNeT++ for simulating computer networks.			
5	Aim: To simulate data transfer between two network nodes using simulation			
6	Aim: To simulate HTTP communication between client and server using socket programming.			
7	Aim: To simulate message queues as an inter-process communication mechanism			
8	Aim: To simulate and analyze routing paths and protocols in a network.			
9	Aim: To simulate different data link layer framing techniques using programming.			
10	Aim: To simulate and implement CRC for error detection			

Signature of Faculty

Course Outcomes

After successful completion of laboratory course, the students will able to:

SN	Outcomes	BT Level	PO
1	Identify and analyze different types of network topologies, devices, and connections.	2	1
2	Configure IP addressing schemes and use basic networking commands for connectivity testing.	3	1, 2
3	Simulate and analyze communication between networked systems using simulation tools like OMNeT++ and Packet Tracer.	3	1, 2, 3
4	Implement and demonstrate data link layer mechanisms such as framing and error detection.	4	1, 2
5	Demonstrate network protocols and routing mechanisms using simulations and socket programming.	3	1, 2, 3
6	Configure VLANs and understand the role of logical segmentation in network design.	3	1, 2, 3

I. Rubrics used for continuous Assessment in every lab session

Skills	Allocated Marks	High	Medium	Low
		10-08	07-04	03-00
Process Related Skills	10	Student performed or executed experiments, obtained results, and drew conclusions fully as per expectation within a due date.	Student performed or executed experiments, obtained results, and drew conclusions partially as per expectation after a due date.	Student performed or executed experiments, obtained results, and drew conclusions below the expectation after two weeks of a due date.
Product Related Skills	10	10-08	07-04	03-01
		Records submitted by the student on due date and found highly satisfactory after evaluation.	Records submitted by the student after one week of due date and found moderately satisfactory after evaluation.	Records submitted by the student after two week of due date and found highly dissatisfactory after evaluation.
Viva Voce	05	05-04	03-02	01
		Student answered the Questions Extraordinarily.	Student answered the Qaestions satisfactorily.	Student answrer the Questions moderately satisfaetorily.
Total Marks for Lab Performance 25 Marks				

II. Rubrics used for External Assessment in Lab Course

Skills	Allocated Marks	Parameters	High	Medium	Low
Process Related Skills	**15**	Handle equipment/ tools/ commands correctly or logic formation and Work cohesively in team **(7)**	Most satisfactory **(07-06)**	Partially successful **(05-04)**	Below expectation **(03-01)**
		Integrate system correctly & measure parameters correctly or Debugging Ability **(8)**	Highly satisfactory **(08-07)**	Partially correct **(05-06)**	Incorrect or unsatisfactory **(04-01)**
Product Related Skills	**10**	Obtain correct results, Answer practical related questions **(5)**	Exceptional **(05-04)**	Satisfactory **(03)**	Unsatisfactory **(02-01)**
		Interpret results, draw conclusion **(5)**	Exceptional **(05-04)**	Satisfactory **(03)**	Unsatisfactory **(02-01)**
Total Marks		**25 Marks**			

PRACTICAL NO. 1

TITLE: Identify & analyze the network topology implemented in the computer network laboratory (consisting of Nodes, Cables, Peripherals & Host Computers)

AIM: To study and identify the physical topology and hardware components of a LAN

OBJECTIVES:

1. Identify various network devices and cables.
2. Understand different topologies used in networking.

SOFTWARE: Cisco Packet Tracer 8.2.2.

THEORY:

To identify the network topology in a computer network laboratory, you can analyze the cables, nodes, peripherals, and host computers.

Steps to Implement and Analyze Network Topology in Cisco Packet Tracer:

1. Launch Cisco Packet Tracer:

Open the Cisco Packet Tracer on your computer.

2. Create the Network Topology:

Start by creating a simple network topology using various devices in Cisco Packet Tracer.

The common elements to include are:

- Host Computers
- Switches
- Routers
- Peripherals (e.g., printers, servers)
- Cables (Ethernet cables, Fiber cables, etc.)

A. Place the Devices:

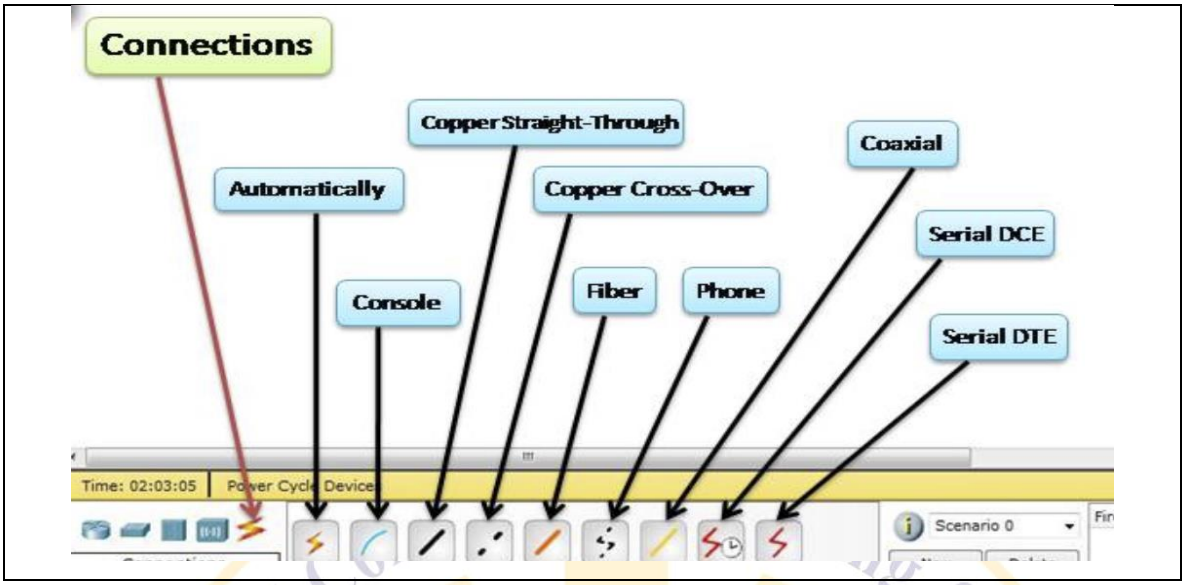
In the Devices Panel on the lower left, you'll find categories like End Devices, Network Devices, and Connection Devices. Use these to place the devices.

- **Host Computers (PCs):** Drag and drop PCs or Laptops to the workspace.
- **Switches:** Drag a Switch to connect multiple devices.
- **Router:** If necessary, add a Router to connect multiple networks.
- **Peripherals:** For example, add a Printer or Server.
- **Wireless Devices:** If using wireless connections, add Wireless Routers or Wireless Access Points.

B. Connect the Devices Using Cables:

Select the appropriate cable:

- **Copper Straight-Through Cable:** Used to connect PCs to switches or routers.
- **Copper Cross-Over Cable**: Used for switch-to-switch or PC-to-PC connections.
- **Fiber Optic Cable:** Used for high-speed connections between routers or switches over long distances.
- **Wi-Fi connections:** If you use wireless devices, configure them with a wireless connection.



Click on a device (e.g., PC or switch) and select the appropriate cable from the Connections options, then connect the devices accordingly.

3. Assign IP Addresses and Configure Devices:

Host Computers: Assign IP addresses to each PC in the network:
Click on a PC → Desktop → IP Configuration → Assign a static IP (e.g., 192.168.1.1, 192.168.1.2, etc.).

Router: Assign IP addresses to the router interfaces for communication between subnets. Select the Router, then access the CLI (Command Line Interface) and configure the router interfaces using commands like:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1  255.255.255.0
Router(config-if)# no shutdown
```

Switches: Typically, do not require IP addresses for Layer 2 (data link layer) switches unless configuring management IPs for Layer 3 switches.

4. Analyze and Identify Network Topology:

Identify the Nodes: In Packet Tracer, each device (PC, router, switch, server, printer) is a node in the network.

Analyze the Cables: Observe the types of cables used (straight-through or crossover) and identify any wireless connections.

Network Devices: Identify the roles of each network device, e.g., a router for routing between subnets, a switch for local connectivity, etc.

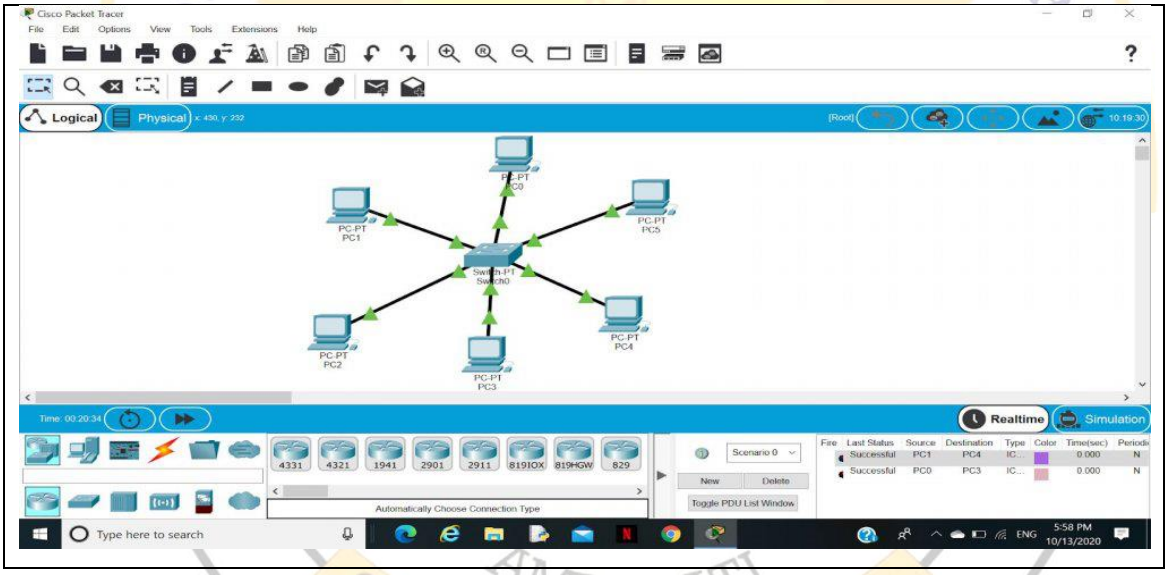
Topology Type: Based on the arrangement, determine the network topology. Some common topologies to consider are:

Star Topology: All devices connect to a central switch.

Bus Topology: All devices are connected to a single central cable.

Mesh Topology: Devices are directly interconnected, providing redundancy.

Hybrid Topology: A combination of multiple topologies.



5. Analyze the Network Components:

- **Host Computers (PCs):** These are end-user devices used for communication and accessing shared resources.
- **Switches:** These connect multiple devices in a local network, forwarding data based on MAC addresses.
- **Router:** Routes data between different subnets or networks, providing inter-network communication.
- **Peripherals:** Printers, servers, and other devices that provide shared resources to the

network.

- **Cables:** Assess the type and functionality of cables used in the network. For example, Ethernet cables can carry data at different speeds (e.g., Cat5e for 100Mbps, Cat6 for 1Gbps).

6. Test the Network and Troubleshoot:

- Use Simulation Mode in Packet Tracer to test network traffic and visualize how data is transferred between devices.
- Check the status of devices, ensure there are no connection issues, and troubleshoot if necessary (e.g., incorrect IP configuration or cabling issues).

CONCLUSION:

_____

_____.

LAB QUESTIONS:

Sr No.	Questions
1	What type of network topology is implemented in the lab? _____ _____ _____
2	What are the advantages and disadvantages of the topology? _____ _____ _____ _____ _____ _____

3	What types of cables are used to connect the devices in the network? _____ _____ _____ _____
4	What is the function of the router, and switch network device? _____ _____ _____ _____ _____ _____ _____ _____
5	Are straight-through or crossover cables used in the network, and for which connections? _____ _____ _____ _____ _____

Signature of Faculty

PRACTICAL NO. 2

TITLE: Perform network IP address configuration (classification of address, static and dynamic address)

AIM: To configure IP addresses for network nodes and verify connectivity.

OBJECTIVES:

1. Learn about IP address classes and subnetting.
2. Configure static and dynamic IPs in a network.

SOFTWARE: Cisco Packet Tracer 8.2.2

THEORY:

To perform network IP address configuration in Cisco Packet Tracer, we typically configure static or dynamic IP addresses on various network devices like routers, switches, and PCs.

1) CLASSIFICATIONS OF IP ADDRESSES

IP addresses are classified into various categories based on their ranges. The most commonly used classifications are:

Public IP addresses: These are globally routable and are assigned to devices on the internet.

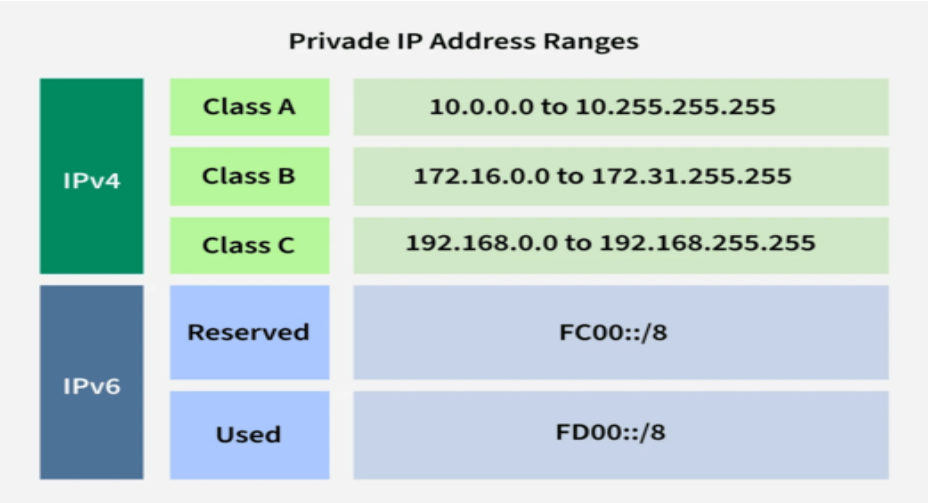
Private IP addresses: These are used within private networks and cannot be routed over the internet.

They are defined by the following ranges:

Class A: 10.0.0.0 - 10.255.255.255

Class B: 172.16.0.0 - 172.31.255.255

Class C: 192.168.0.0 - 192.168.255.255



Privade IP Address Ranges

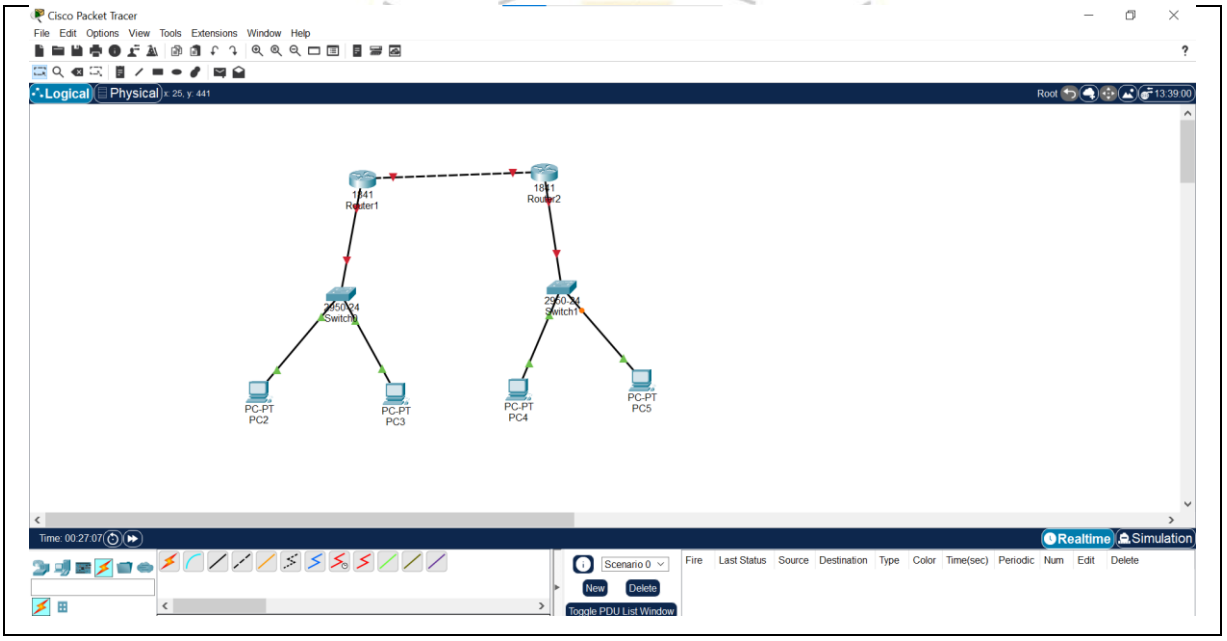
IPv4	Class A	10.0.0.0 to 10.255.255.255
	Class B	172.16.0.0 to 172.31.255.255
	Class C	192.168.0.0 to 192.168.255.255
IPv6	Reserved	FC00::/8
	Used	FD00::/8

2) STATIC IP ADDRESS CONFIGURATION

1. Open Cisco Packet Tracer: Launch Cisco Packet Tracer on your computer.

2. Create the Network:

- Drag and drop devices from the left panel (such as PCs, routers, or switches) onto the workspace.
- Connect the devices using cables (e.g., Copper Straight-Through cable) by clicking on one device's interface and then connecting it to another device's interface.



3. **Select the PC (or device):** Click on the PC (or device) to open its configuration options.

4. Open Desktop Tab

- Inside the PC configuration window, click the "Desktop" tab.

5. Click on IP Configuration

- Now, click on the "IP Configuration" option. This window will allow you to manually set the IP address, subnet mask, and default gateway.

6. Assign Static IP Address

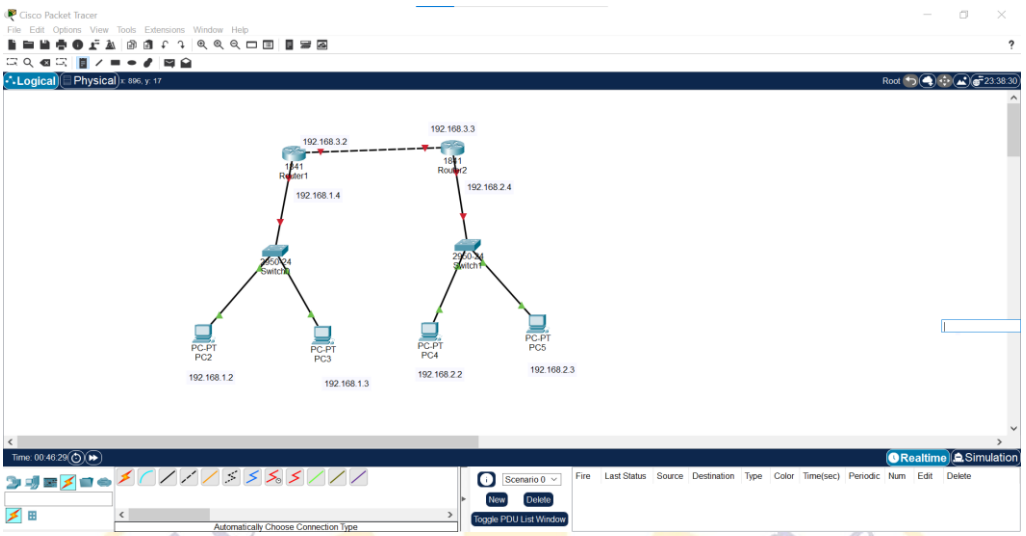
- **IP Address**: In the "IP Address" field, enter the static IP address you want to assign to the device. For example, 192.168.1.10.
- **Subnet Mask**: The subnet mask field will auto-populate with the default value (255.255.255.0) if you're using a typical home network. You can leave this as it is, or adjust it based on your network's requirements.
- **Default Gateway**: Enter the default gateway IP address (usually the router's IP address). For example, 192.168.1.1.

7. Test the Configuration

- To check if the static IP address is configured properly, you can use the "Command Prompt" (found in the "Desktop" tab) and use the ping command to test connectivity:
    - Ping the default gateway: ping 192.168.1.1
    - Ping another device in the same network (if applicable).

8. Save the Configuration

- You can save the configuration by simply saving the Cisco Packet Tracer project or noting down the IP settings for reference.



3) DYNAMIC IP ADDRESS CONFIGURATION (DHCP)

Here's a step-by-step guide to configuring Dynamic IP Address (DHCP) in Cisco Packet Tracer:

1. Open the Cisco Packet Tracer

Launch the Cisco Packet Tracer on your computer.

2. Create the Network

Create a basic network topology by dragging and dropping devices (e.g., PCs, routers, switches) from the left panel onto the workspace.

3. Add a Router (for DHCP)

You need a router to act as the DHCP server. Drag a router (e.g., Cisco 1941 or any model) onto the workspace.

4. Configure the Router for DHCP

- Click on the router to open its configuration window.
- Click the "CLI" tab to access the router's command-line interface.

Step 1: Enter Global Configuration Mode

```
Router> enable

Router# configure terminal

Router(config)#
```

Step 2: Define the DHCP Pool

Create a DHCP pool by defining the network range for IP addresses that the router can assign to devices.

```
Router(config)# ip dhcp pool MYPOOL

Router(dhcp-config)# network 192.168.1.0 255.255.255.0

Router(dhcp-config)# default-router 192.168.1.1

Router(dhcp-config)# dns-server 8.8.8.8

Router(dhcp-config)# exit
```

Step 3: Exclude IP Addresses (Optional)

If you want to exclude certain IP addresses from the DHCP pool (e.g., to reserve them for static IPs), use the ip dhcp excluded-address command.

```
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

This will reserve IP addresses from 192.168.1.1 to 192.168.1.10 for static configuration.

5. Configure the Router's Interface

Configure the router's interface with a static IP address, which will act as the default gateway for the network.

```
Router(config)# interface gigabitEthernet 0/0

Router(config-if)# ip address 192.168.1.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit
```

6. Connect Devices to the Router

Connect a switch or directly connect a PC to the router's interface using appropriate cables (usually copper straight-through).

7. Configure PCs to Use DHCP

- Click on a PC in the network to open its configuration window.
- Go to the Desktop tab and then click on IP Configuration.
- Select DHCP instead of manually setting the IP address.
- The PC will now request an IP address from the router's DHCP server.

Note:

Once the PC requests an IP from the DHCP server, it should receive an IP in the range of 192.168.1.11 to 192.168.1.254, depending on how you've set up your DHCP pool.

8. Verify DHCP Configuration

To verify the DHCP configuration, you can check the IP address assigned to the PC by going to IP Configuration on the PC. It should display an IP from the range you set.

You can also use the show ip dhcp binding command on the router to check the IP addresses that have been assigned:

```
Router# show ip dhcp binding
```

9. Test Connectivity

Use the Command Prompt on the PC (found under the Desktop tab) and test connectivity by

pinging the default gateway or another device on the network:

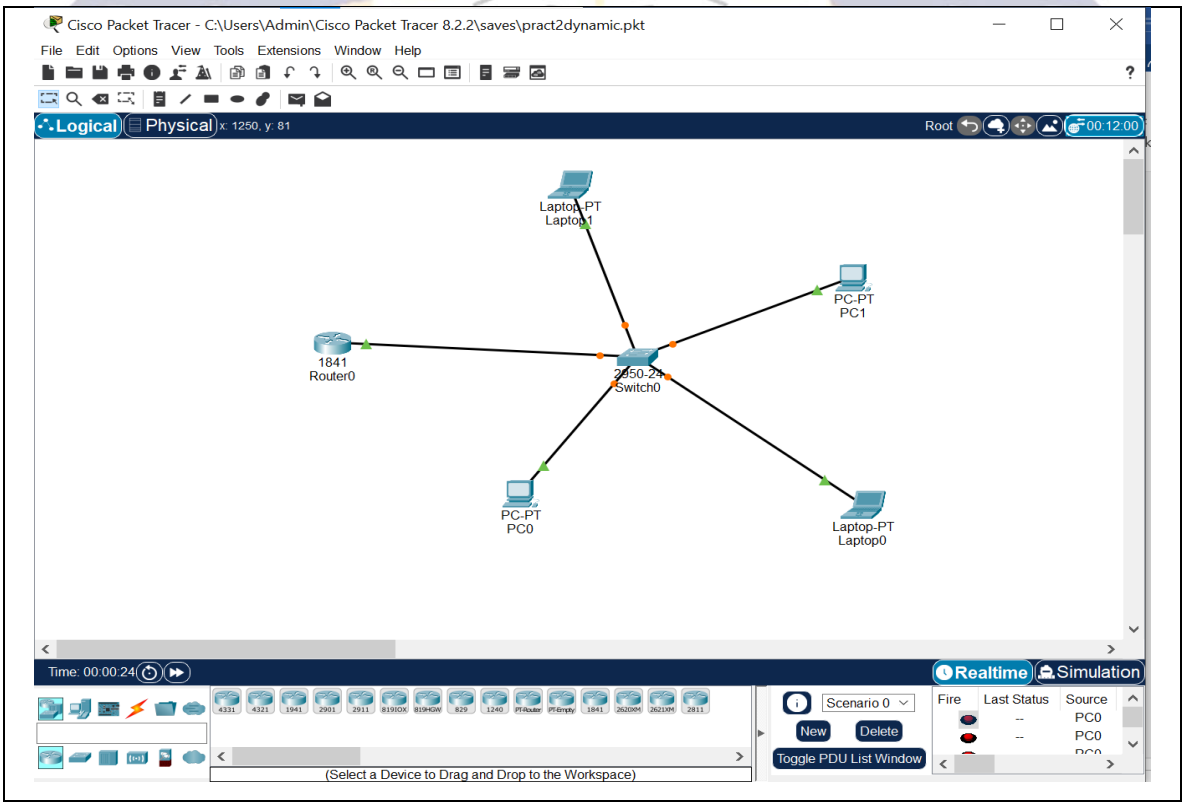
ping 192.168.1.1 (default gateway).

Example:

- **DHCP Pool:** 192.168.1.0/24
- **Exclusion Range:** 192.168.1.1 to 192.168.1.10
- **Default Gateway:** 192.168.1.1
- **DNS Server:** 8.8.8.8

DYNAMIC IP CONFIGURATION ON A PC:

1. Add a PC to the workspace.
2. Click on the PC and go to the "Desktop" tab.
3. Select IP Configuration.
4. Select DHCP (Dynamic Host Configuration Protocol).
5. The PC will automatically receive an IP address, subnet mask, and gateway from the router's DHCP server.



Verify Configuration

Once the configuration is done, you can verify the connectivity and IP address configuration using commands:

On the PC (to check IP): Open Command Prompt (from the Desktop tab).

2. Type:

```
ipconfig
```

This will display the PC's IP address, subnet mask, and gateway.

On the Router (to check IP):

1. Enter the router's CLI and type:

```kotlin
show ip interface brief
```

This will display the status of all interfaces, including their IP addresses.

To Test Connectivity:

1. From the PC, use the **ping** command to test connectivity:

```
ping 192.168.1.1
```

If the PC is correctly configured and connected, you should see successful ping replies.

OUTPUT ON SIMULATOR:

CONCLUSION:

_____

_____

_____

LAB QUESTIONS:

Sr No.	Questions
1	What are the three main classes of IP addresses used in private networks? a) Class A, Class B, Class C b) Class A, Class D, Class E c) Class B, Class C, Class D d) Class A, Class B, Class D _____ _____ _____
2	Which of the following IP address ranges is part of the Class B private IP range? a) 192.168.1.0 - 192.168.255.255 b) 10.0.0.0 - 10.255.255.255 c) 172.16.0.0 - 172.31.255.255 d) 127.0.0.0 - 127.255.255.255 _____ _____ _____

3	Which IP address configuration method assigns an IP address dynamically from a pool of available addresses? a) Static IP addressing b) Dynamic IP addressing (DHCP) c) NAT (Network Address Translation) d) Subnetting _____ _____
4	Explain the difference between static and Dynamic IP Address. _____ _____ _____ _____
5	When configuring a static IP address on a router's interface, which of the following commands must be executed? a) ip address <IP address> <subnet mask> b) dhcp pool <pool-name> c) ip route <destination> <subnet mask> <next hop> d) ip address dhcp _____ _____ _____

Signature of Faculty

PRACTICAL NO. 3

TITLE: Execute basic networking commands like ping, traceroute, netstat, ipconfig, nslookup

AIM: To execute common network diagnostic commands and interpret their results. Objectives:

OBJECTIVES:

1. Use commands to check connectivity and routing.
2. Analyze network configuration and performance.

SOFTWARE: Cisco Packet Tracer 8.2.2

THEORY:

In Cisco Packet Tracer, you can execute a variety of basic networking commands to troubleshoot and verify network configurations. Below is a step-by-step guide on how to use the following basic networking commands: ping, traceroute, netstat, ipconfig, and nslookup.

1. PING COMMAND

The ping command is used to test the connectivity between two devices on a network.

Example: You want to test the connectivity between a PC and a router.

Steps:

1. Drag and drop a PC onto the workspace in Cisco Packet Tracer.
2. Click on the PC to open the configuration window.
3. Go to the Desktop tab and select Command Prompt.
4. Type the following command and press Enter:

```
ping <destination IP address>
```

For example, if you want to ping a router's IP:

```
ping 192.168.1.1
```

This command sends ICMP Echo Request messages to the specified IP address and waits for replies. It will show you if the destination is reachable or if there is packet loss.

2. TRACEROUTE COMMAND

Trace the route packets take to reach the destination.

Steps:

1. Follow steps 1-3 from the Ping command to access the Command Prompt of the PC.
2. Type the following command:

```
traceroute <destination IP address>
```

Example:

```
traceroute 192.168.1.1
```

The traceroute command shows you the intermediate devices (routers) the packets go through before reaching the destination. This helps in identifying network bottlenecks or routing issues.

3. NETSTAT COMMAND

Display active network connections and network statistics.

Steps:

1. Follow steps 1-3 from the Ping command to access the Command Prompt.
2. Type the following command:

```
Netstat
```

This will display active network connections and listening ports. You will see information like local and remote addresses, connection status, and protocol type (TCP or UDP).

4. IPCONFIG COMMAND

View the device's current IP configuration.

Steps:

1. Follow steps 1-3 from the Ping command to access the Command Prompt.
2. Type the following command:

```
Ipconfig
```

This will show you the IP address, subnet mask, default gateway, and other network interface information of the PC.

If you're working with a router, you can check the router's IP configuration using:

- Click on the router.
- Go to the CLI tab and enter:

```
show ip interface brief
```

5. NSLOOKUP COMMAND

Resolve domain names to IP addresses and troubleshoot DNS issues.

Steps:

1. Follow steps 1-3 from the Ping command to access the Command Prompt.
2. Type the following command:

```
nslookup <domain name>
```

Example:

```
nslookup www.google.com
```

This command will query the DNS server and resolve the domain name (e.g., www.google.com) into an IP address.

Note: For nslookup to work, you must have a DNS server configured, or it will use a default DNS server.

EXAMPLE NETWORK TOPOLOGY FOR TESTING:

1. **Add devices:**

- A PC with IP address 192.168.1.10.
- A Router with IP address 192.168.1.1 on its interface connected to the PC.

2. **Connect the devices:**

Use a Copper Straight-Through cable to connect the PC to the router's interface.

3. **Test Commands:**

On the PC, open the Command Prompt and run the following commands:

- ping 192.168.1.1 (ping the router).
- traceroute 192.168.1.1 (trace the route to the router).
- netstat (see the active connections).
- ipconfig (view the IP configuration).
- nslookup www.google.com (resolve a domain name).

OUTPUT ON SIMULATOR:

CONCLUSION:

_____

_____.

LAB QUESTIONS:

Sr No.	Questions
1	What does the ping command do, and how can it help troubleshoot network issues? _____ _____ _____ _____
2	What is the purpose of the traceroute command? _____ _____ _____ _____
3	How can you use the netstat command to identify active connections on a device? _____ _____ _____ _____
4	What information does the ipconfig command display on a PC? _____ _____ _____ _____

5	What is the function of the nslookup command? _____ _____ _____ _____

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PRACTICAL NO. 4

Title: Install and configure the OMNeT++ simulation tool

Aim: To install and configure OMNeT++ for simulating computer networks.

Objectives:

1. Learn setup and basic features of OMNeT++.

2. Simulate basic data communication models.

SOFTWARE: OMNeT++ 5.1

THEORY :

OMNeT++ is an extensible, modular, component-based C++ simulation library and framework, primarily for building network simulators. "Network" is meant in a broader sense that includes wired and wireless communication networks, on-chip networks, queueing networks, and so on. Domain-specific functionality such as support for sensor networks, wireless ad-hoc networks, Internet protocols, performance modeling, photonic networks, etc., is provided by model frameworks, developed as independent projects. OMNeT++ offers an Eclipse-based IDE, a graphical runtime environment, and a host of other tools.

OMNeT++ provides a component architecture for models. Components (modules) are programmed in C++, then assembled into larger components and models using a high-level language (NED). Reusability of models comes for free. OMNeT++ has extensive GUI support, and due to its modular architecture, the simulation kernel (and models) can be embedded easily into your applications.

Components

1. The main ingredients of OMNeT++ are:
2. Simulation kernel library (C++)
3. The NED topology description language

4. Simulation IDE based on the Eclipse platform
5. Interactive simulation runtime GUI (Qtenv)
6. Command-line interface for simulation execution (Cmdenv)
7. Utilities (makefile creation tool, etc.)
8. Documentation, sample simulations, etc.

SIMULATION IDE

The Simulation IDE is a powerful, feature-rich environment for developing, running, and evaluating simulation models. Please note that using the IDE is entirely optional; model development and execution are also possible and fully supported through the command line or alternative editors/IDEs.

INSTALL AND CONFIGURE THE OMNET++ :

Installing and configuring OMNeT++ involves several steps. Below is a guide to help you through the process. Please ensure you have the necessary permissions and system requirements before proceeding.

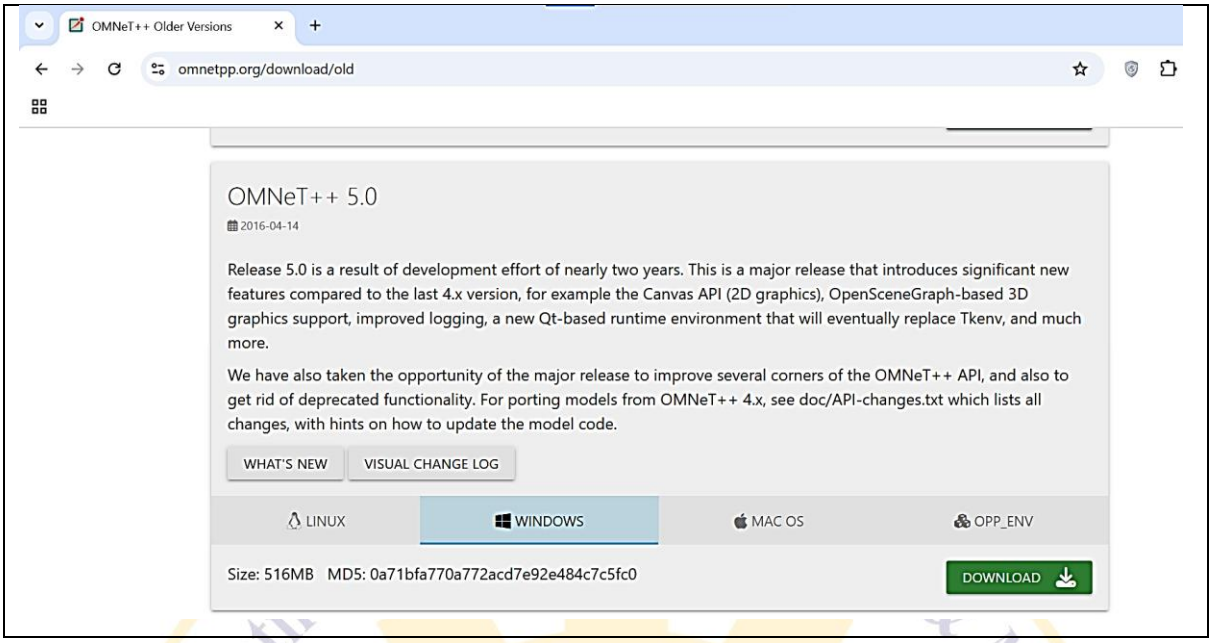
1) SYSTEM REQUIREMENT

To perform the installation process, we need the following requirements,
1) OS: Windows 10 pro
2) Install Omnet++ 5.1

2) DOWNLOAD THE OMNET++ PACKAGE

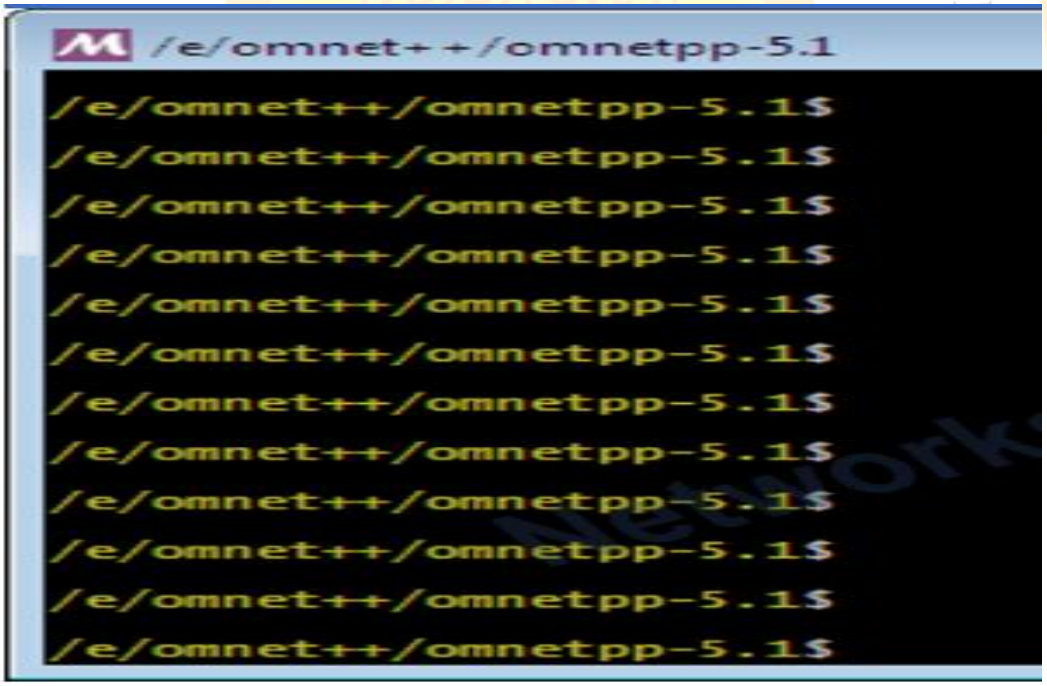
Download the omnet++ package from the following url, in the site there multiple verion of the omnet++ is listed. Select your needed version of omnet++, here we select the eversion omnet++ 5.1 , after that click the download button , which is displayed in the below of selected version

https://omnetpp.org/download/old.html



3) OPEN THE TERMINAL/ MINGWENV WINDOW

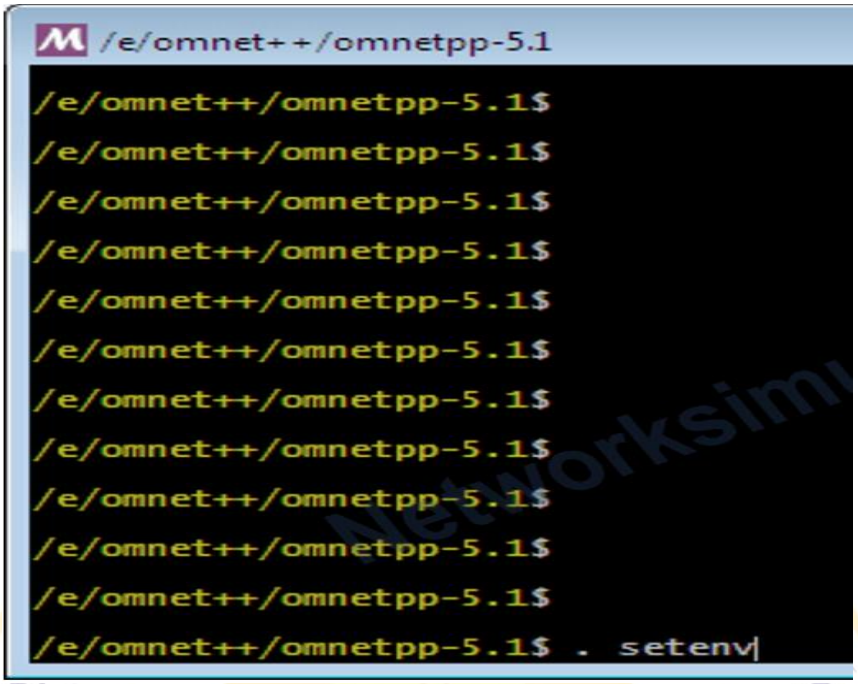
Open the mingwenv window. By double click mingwenv window from the omnet++ installed location.



4) EXECUTE THE COMMAND WINDOW

In the mingwenv command window ,

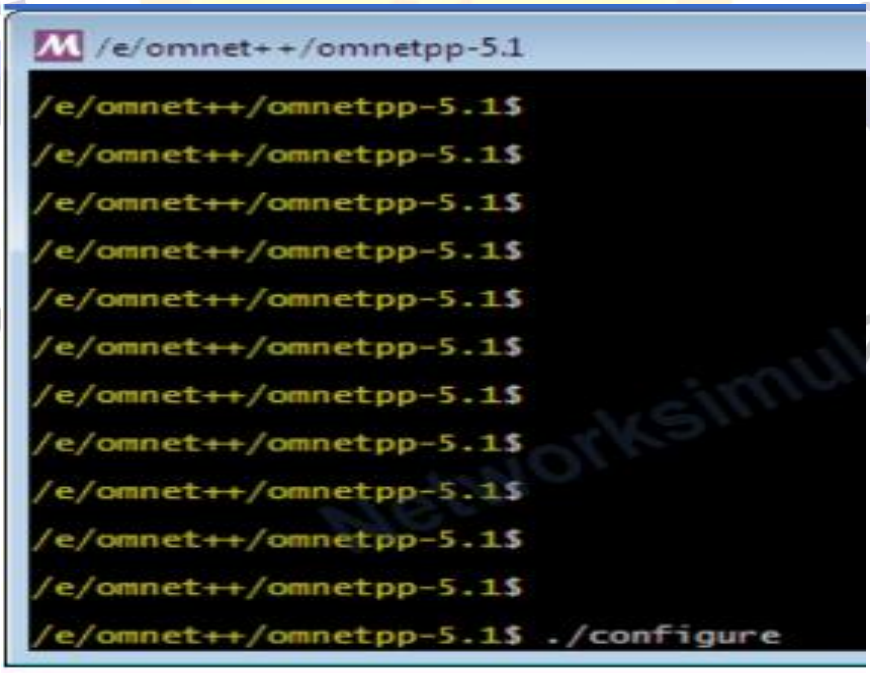
Execute the command -> **. setenv**



5) EXECUTE THE CONFIGURE COMMAND WINDOW

In the mingwenv command window ,

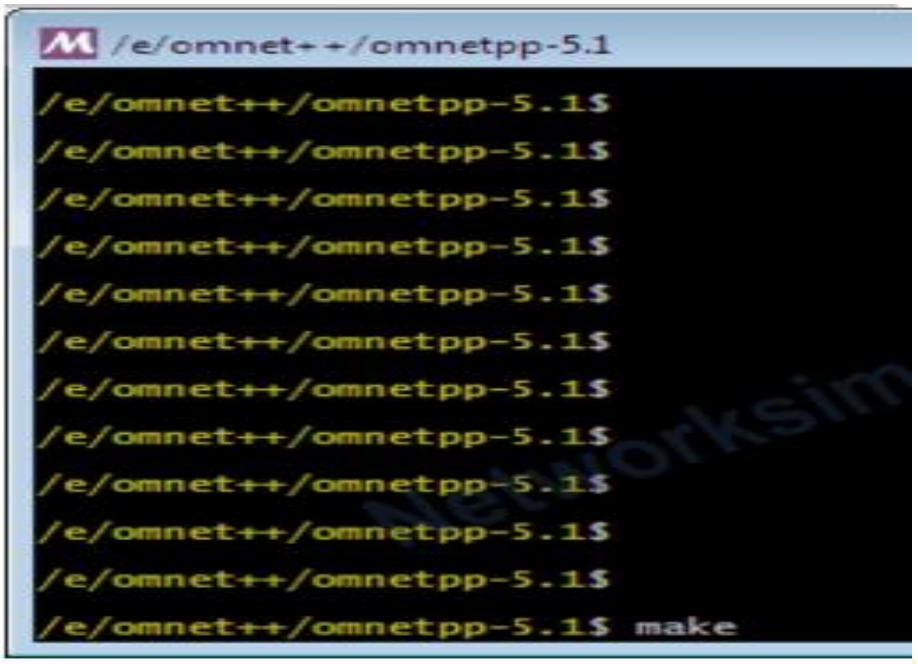
Execute the command -> **./configure**



6) EXECUTE THE MAKE COMMAND WINDOW

In the mingwenv command window,

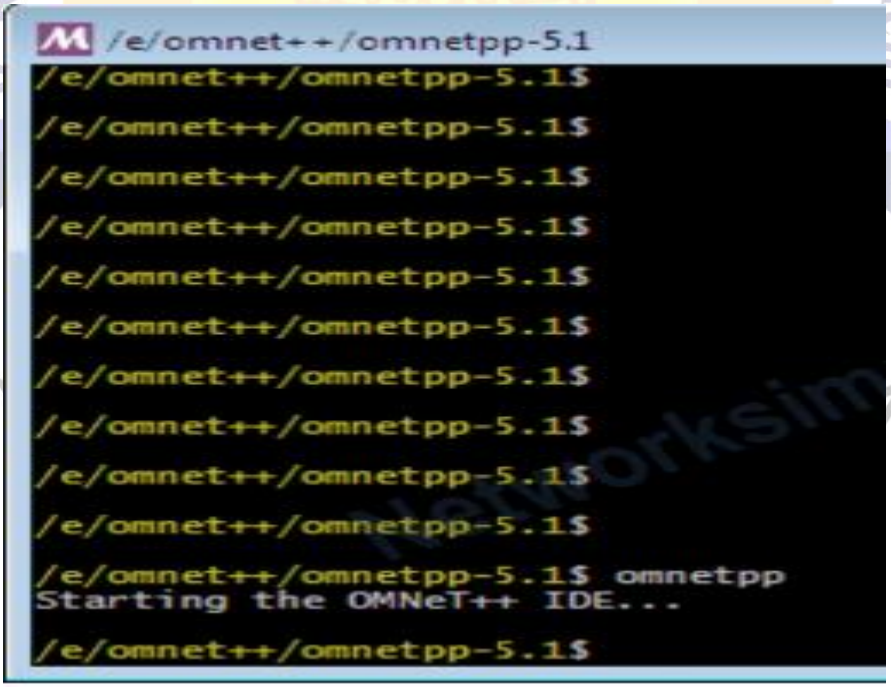
Execute the command -> **make**

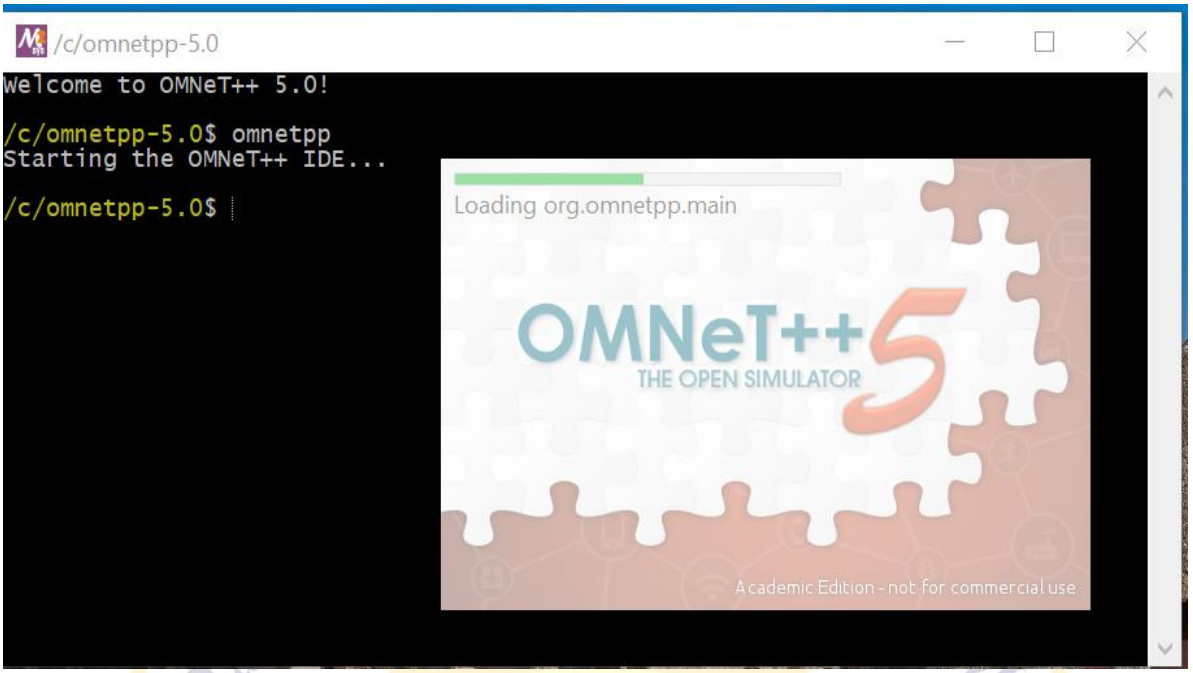


7) EXECUTE THE OMNET++ COMMAND WINDOW

In the mingwenv command window ,

Execute the command -> omnetpp





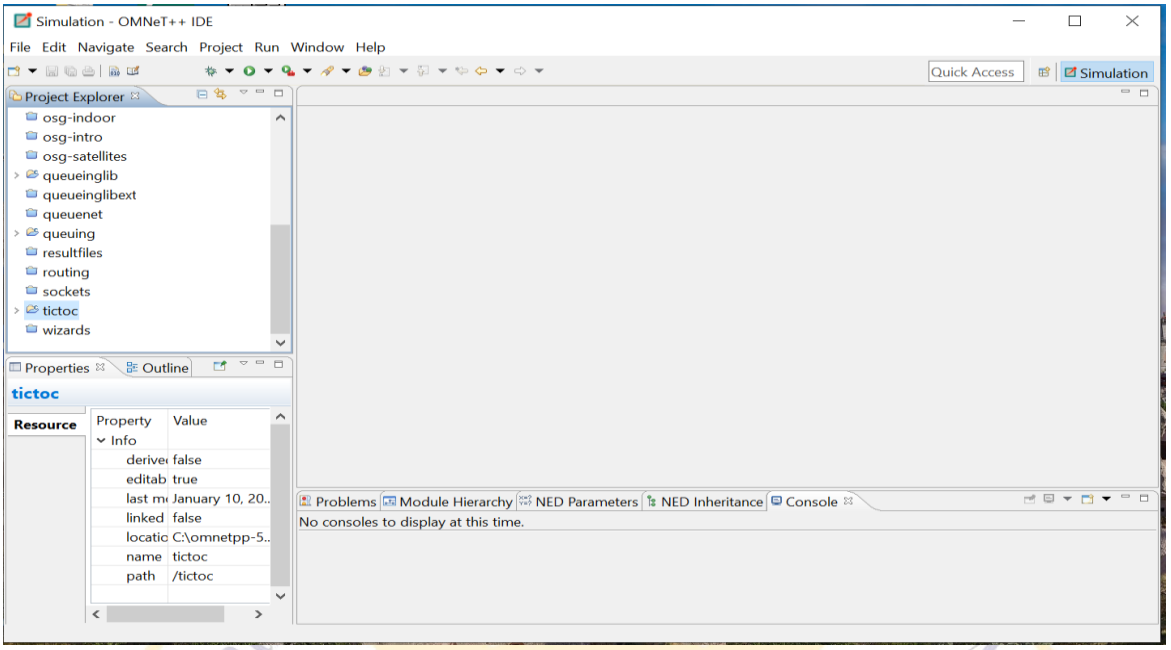
8) SELECT THE WORKSPACE WINDOW

Select the stored workspace, with full location, to select the workspace, use the browse button and select the location, and press the ok button or type the full location. , like the example below

9) LOADED WORKSPACE WINDOW

By use the browse button and select the location and press the ok button or type the full location. , like the example below,

View the loaded workspace window



OUTPUT ON SIMULATOR:

CONCLUSION:

_____

_____.

LAB QUESTIONS:

Sr No.	Questions
1	What is OMNeT++ and what are its primary applications? _____ _____ _____ _____

2	What are the system requirements for installing OMNeT++? _____ _____ _____
3	What steps are involved in downloading OMNeT++? _____ _____ _____
4	What command is used to extract the downloaded OMNeT++ archive? _____ _____ _____
5	What is the purpose of the 'make' command in the installation process? _____ _____ _____

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PRACTICAL NO. 5

TITLE: Demonstrate and simulate data communication in a simple network

AIM: To simulate data transfer between two network nodes using simulation

OBJECTIVES:

1. Observe data packet flow between devices.

2. Analyze communication at various OSI layers

SOFTWARE: OMNeT++ 5.1

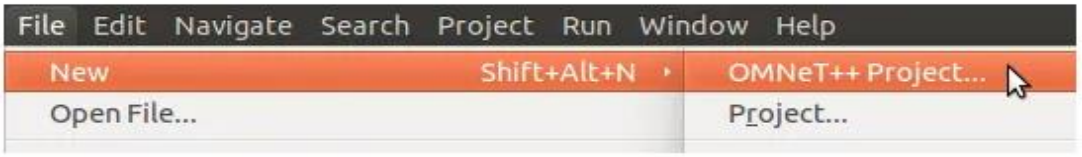
THEORY:

OMNeT++ is an extensible, modular, component-based C++ simulation library and framework, primarily for building network simulators. "Network" is meant in a broader sense that includes wired and wireless communication networks, on-chip networks, queueing networks, and so on.

STEP 1: SET UP OMNET++

1. Install OMNeT++: Ensure you have OMNeT++ installed on your system. You can download it from the OMNeT++ website.
2. Create a New Project: Open OMNeT++ and create a new project by selecting

   File > New > OMNeT++ Project



STEP 2: DEFINE THE NETWORK TOPOLOGY (NED FILE)

In OMNeT++, the network topology is defined using NED files. This file describes the components, their interconnections, and communication protocols.

1. Navigate to the src directory in your project.
2. Create a new NED file, e.g., My_hello_world.ned.
3. Define the nodes and connections

This creates a network with two nodes, node1 and node2, connected in a simple bi-directional way.

STEP 3: DEFINE THE MESSAGE GENERATION (C++ FILE)

Now, let's write the logic that will handle the communication between the nodes. We'll use C++ to implement the behaviour of the nodes.

1. Create a new C++ file named SimpleNode.cc in the src folder.
2. Implement the behaviour to send and receive messages.

STEP 4: CONFIGURE THE SIMULATION

Create a Configuration File: In your project directory, create a file named **omnetpp.ini** to configure the simulation.

STEP 5: BUILD AND RUN THE SIMULATION

1. **Build the Project:** In OMNeT++, right-click on your project and select Build Project
2. **Run the Simulation:** After building, right-click on the project again and select **Run As > OMNeT++ Simulation**

You should see the simulation running, and the output will display the messages being sent and received between the nodes.

SOURCE CODE

1) .ned file

2) <u>.cc source file</u>

<u>OUTPUT:</u>

1) <u>DESIGN NETWORK:</u>

2) OUTPUT ON THE SIMULATIOR

CONCLUSION:

_____

_____.

LAB QUESTIONS:

Sr No.	Questions
1	What is data communication, and why is it important in a network? _____ _____ _____
2	What are the key components of a simple network? _____ _____ _____
3	What is a .ned file? _____ _____ _____

4	What is the purpose of a .cc file in omnet++? _____ _____ _____
5	How to run the simulation? _____ _____ _____

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PRACTICAL NO. 6

TITLE: Demonstrate the simulation of Hypertext Transfer Protocol (HTTP) using network sockets

AIM: To simulate HTTP communication between client and server using socket programming.

OBJECTIVES:

1. Understand client-server interaction over HTTP.
2. Implement a simple HTTP request–response model.

SOFTWARE: Omnet++ 5.1

THEORY:

STEPS TO DEMONSTRATE MESSAGE QUEUES IN AN INTERMEDIATE NODE IN OMNET++

Simulating the Hypertext Transfer Protocol (HTTP) using network sockets in OMNeT++ requires several steps to set up the components involved in this protocol. Here's a step-by-step outline of how to approach simulating HTTP communication using OMNeT++:

1) CREATE A NEW OMNET++ PROJECT

Once OMNeT++ is installed, create a new project in OMNeT++ for the HTTP simulation.

- Open OMNeT++ IDE.
- Select File -> New -> OMNeT++ Project.
- Enter the project name (e.g., HTTP_Simulation) and click Finish.

2) DEFINE THE NETWORK TOPOLOGY

Create the necessary components (e.g., HTTP client, HTTP server, and network links) for your simulation. In OMNeT++, you define network topology in the .ned files.

Example **HTTP_Simulation.ned.**

In this example, the client and server modules are defined, and a TCP connection exists between them.

3) CREATE MODULES FOR HTTP CLIENT AND SERVER

In OMNeT++, you can model HTTP clients and servers using simple TCP models or using TCPApp and TCPServer modules. You need to extend these modules or use simple application models that simulate HTTP behavior.

- **Example TCP Client Module:**
  Create a new module for the HTTP client (e.g., TCPClient): TCPClient.cc (C++ code)

- **Example TCP Server Module:**
  Create a new module for the HTTP server (e.g., TCPServer):

- **TCPServer.cc (C++ code)**

4) SET UP NETWORK LAYER AND APPLICATION LAYER

1. OMNeT++ allows you to simulate different layers of the protocol stack. For HTTP over TCP, you need to ensure that the TCP protocol stack is properly configured.

2. Use the TCP module for transport layer communication.
3. Optionally, use the TCPApp module for application-level HTTP communication.
4. You will need to configure the parameters in the .ini file for things like packet size, simulation time, and message interval. **Example omnetpp.ini**

5) BUILD AND RUN THE SIMULATION

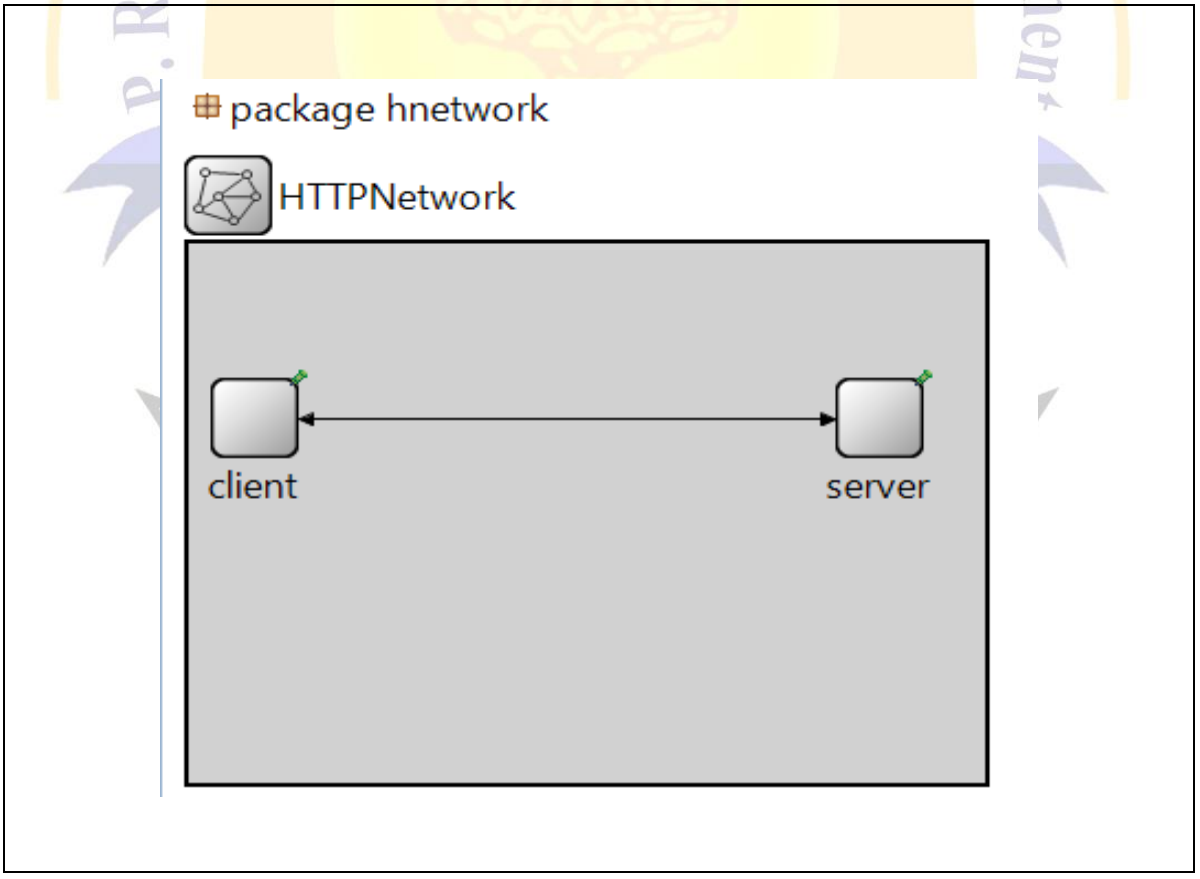
After creating the models and configurations, build and run the simulation:

1. In OMNeT++, click Build to compile your project.
2. Click Run to start the simulation. You can observe the communication between the client and server in the simulation environment.

6) OBSERVE HTTP REQUEST/RESPONSE COMMUNICATION

1. Once the simulation is running, you can check the interactions between the client and server:
2. Check the output logs for HTTP requests and responses.
3. Use OMNeT++'s Analyzer or Inspector to view the packet flow and network interactions.
4. Optionally, visualize the communication process with OMNeT++'s GUI features (e.g., showing messages and connections in the network).

DESIGN NETWORK:



SOURCE CODE:

1) .**HTTPNetwork.ned**

2) **HTTPServer.ned**

3) **HTTP Client.ned**

4) **HTTPServer.cc**

5) <u>HTTPClient.cc</u>

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6) omnetpp.ini

OUTPUT ON SIMULATIOR:

CONCLUSION:

_____

_____

_____

LAB QUESTIONS:

Sr No.	Questions
1	What components are necessary to simulate HTTP communication in OMNeT++? _____ _____ _____ _____ _____
2	What are HTTP Request Messages? _____ _____ _____
3	On which port http works? _____ _____ _____

4	Explain HTTP client module. _____ _____ _____
5	What is HTTP explain working? _____ _____ _____

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PRACTICAL NO. 7

TITLE: Demonstrate the working of message queues available at an intermediate node

AIM: To simulate message queues as an inter-process communication mechanism.

OBJECTIVES:

1. Understand how message queues manage data.
2. Demonstrate queue operations in networked nodes

SOFTWARE: Omnet++ 5.1

THEORY:

In OMNeT++, message queues are used to manage messages that need to be processed by a module, especially for modules such as routers, gateways, or intermediate nodes in a network. These queues ensure that messages are processed in the order they arrive or based on some priority if needed.

STEPS TO DEMONSTRATE MESSAGE QUEUES IN AN INTERMEDIATE NODE IN OMNET++

1. CREATE A NEW OMNET++ PROJECT

Open OMNeT++ IDE and create a new project:

File -> New -> OMNeT++ Project

Choose an Empty Project and name it MessageQueueExample.

2. CREATE THE NED NETWORK MODEL

In the src folder, create a file called MessageQueueExample.ned. This file will define the network structure. Here we will define:

Two nodes: Sender (sending messages) and Receiver (receiving messages).

An IntermediateNode that will act as the node where the message queue is implemented.

Sender: Generates and sends messages to the intermediate node.

IntermediateNode: Holds messages in a queue and forwards them to the receiver.

Receiver: Receives messages from the intermediate node.

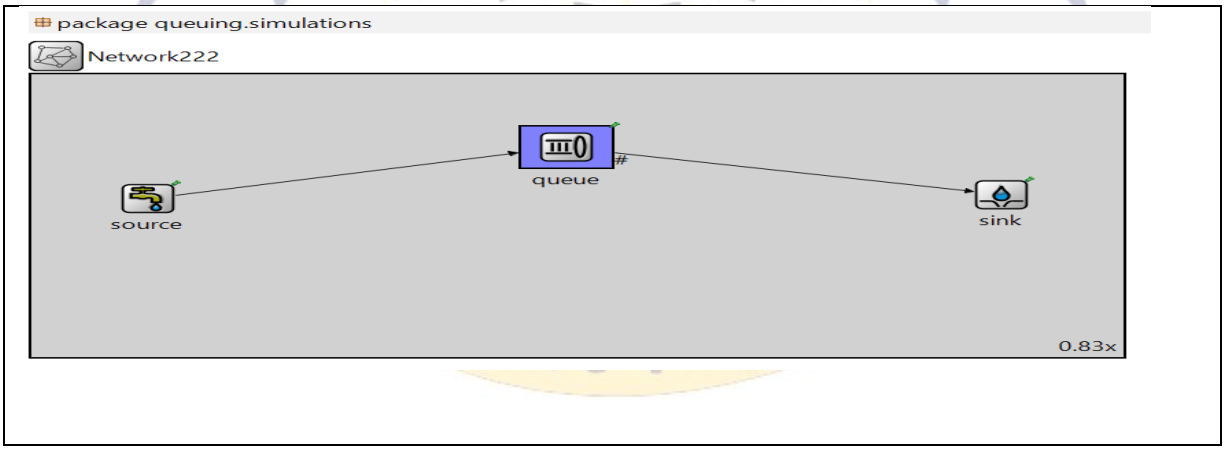
1. Configure Simulation Parameters (omnetpp.ini)

In the omnetpp.ini file, you can set parameters for the simulation such as message generation intervals and the queue length in the intermediate node.

2. Run the Simulation

- After compiling the project, run the simulation. You should observe the following:
- The Sender generates messages every second.
- The IntermediateNode holds messages in its queue (up to the specified maximum length) and forwards them after a small delay.
- The Receiver receives messages forwarded from the IntermediateNode.

DESIGN:



SOURCE CODE:

1) .NED File:

2) .ini File:

OUTPUT ON SIMULATIOR:

CONCLUSION:

_____

_____.

LAB QUESTIONS:

Sr No.	Questions
1	What is a message queue, and how does it function in an intermediate node?
2	What are the benefits of using message queues in an intermediate node?

3	What types of message queues are available? _____ _____ _____ _____ _____
4	How are message queues configured in OMNeT++? _____ _____ _____ _____ _____
5	What happens when a message queue overflows? _____ _____ _____ _____ _____

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PRACTICAL NO. 8

TITLE: Observe simulation to identify various routes data packets may take while traveling from source to destination

AIM: To simulate and analyze routing paths and protocols in a network.

OBJECTIVES:

1. Observe path selection in routing.
2. Compare routing decisions across multiple protocols

Theory

Routing is the process of selecting the best path for data packets to travel from a source to a destination in a network. Routing protocols dynamically build routing tables and determine the route based on metrics such as hop count, cost, bandwidth, delay, or link status.

Types of Routing Protocols

1. Static Routing
   - Manually configured, fixed path
   - No automatic re-routing on link failure
2. Dynamic Routing Protocols
   - Automatically discover and maintain routes
   - Adjust to network changes, failures, and topology updates
     Examples used in simulation:
   - RIP (Routing Information Protocol) – based on hop count
   - OSPF (Open Shortest Path First) – uses link cost (Dijkstra's algorithm)
   - EIGRP (if available) – uses composite metrics

In simulations, multiple routes may be available between source and destination. Routing protocols select the optimal route depending on their metrics and configuration.

Network Topology (Example Used)

Network Topology: Source → Multiple Routers → Destination (Multiple alternate paths available)

- Multiple paths from Source → Destination
- Routing protocols will select different paths depending on cost/hops
- Topology helps analyze routing behavior

Simulation Steps (OMNeT++ / INET Framework)

Step 1: Create a network topology with at least two distinct paths.

Step 2: Configure devices (routers and hosts) with IP addresses.

Step 3: Enable routing protocols:

- Static routing
- RIP
- OSPF

Step 4: Run the simulation and send packets from Source to Destination.

Step 5: Observe:

- Selected path
- Hop-by-hop movement of packets
- Routing table entries

Step 6: Change routing protocol or link cost.

Step 7: Compare which protocol chooses which path and why.

Step 8: Analyze results and record observations.

SOURCE CODE / CONFIGURATION (Example OMNeT++ NED):

```
network RoutingExample
{
        submodules:
        source: StandardHost;
        router1: Router;
        router2: Router;
        router3: Router;
        destination: StandardHost;
        connections:
        source.pppg++ <--> router1.pppg++;
```

```
        router1.pppg++ <--> router2.pppg++;
        router1.pppg++ <--> router3.pppg++;
        router2.pppg++ <--> destination.pppg++;
        router3.pppg++ <--> destination.pppg++;
}
```

Observations

1. Static Routing

   Fixed path selected:
   Example: R1 → R2 → R3
   No re-routing, even if a link becomes slow or costly

2. RIP Routing

   Chooses path with minimum hop count
   Example: If both paths have equal hops, RIP may choose first discovered path.

3. OSPF Routing

   Selects path with lowest total cost
   Uses Dijkstra's shortest-path-first (SPF) algorithm
   More efficient and adjusts dynamically to link changes.

Conclusion

_____

_____.

<u>LAB QUESTIONS:</u>

Sr No.	Questions
1	What is routing and why is it necessary? _____ _____ _____ _____ _____ _____

2	What is the difference between static and dynamic routing?
3	How does RIP select a path?
4	What metric does OSPF use?
5	Why does OSPF react faster to network changes than RIP?

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PRACTICAL NO. 9

TITLE: Implement data link layer framing methods such as Character Count, Character Stuffing, and Bit Stuffing

AIM: To simulate different data link layer framing techniques using programming.
OBJECTIVES:

1. Implement framing methods using C.

2. Understand framing and synchronization concepts

THEORY:

Data link layer :

The **Data Link Layer** is the second layer in the OSI (Open Systems Interconnection) model, and its main job is to provide reliable communication over a physical link by managing frame construction, error detection, and flow control. It takes data from the Network Layer and encapsulates it into frames to be transmitted over the physical medium.

- **Frame Header**: It contains the source and the destination addresses of the frame and the control bytes.
- **Payload field**: It contains the message to be delivered.
- **Trailer**: It contains the error detection and error correction bits. It is also called a Frame Check Sequence (FCS).
- **Flag**: Two flag at the two ends mark the beginning and the end of the frame.

The primary purpose of framing methods like character count, character stuffing, and bit stuffing is to ensure that the data can be correctly interpreted by the receiver, even when transmitted across potentially unreliable or noisy channels.

Character Count: This method involves adding a header to each frame that indicates the

number of characters in the frame. The receiver reads this count to determine how much data is in the frame. The challenge with this method is handling cases where errors cause the receiver to misinterpret the count, so error detection mechanisms are often used alongside it.

Character Stuffing: This technique involves inserting a special character (often an escape character) into the data stream when certain patterns or characters that might otherwise be mistaken for frame delimiters appear in the data itself. This ensures that the delimiter or special marker used to indicate the frame boundary doesn't appear in the actual data, which could confuse the receiver.

Bit Stuffing: Bit stuffing is used when the data is transmitted as a stream of bits. A special bit pattern (e.g., a "flag" sequence like 01111110) is used to signify the start and end of a frame. If the data stream contains five consecutive '1's, a '0' is inserted (stuffed) after them to prevent the data from accidentally resembling the flag pattern.

These methods collectively help manage **data transparency**
—the ability to safely transmit data that could otherwise interfere with the framing mechanism
—while also ensuring the **synchronization** of both sender and receiver.

By handling synchronization and data separation through these framing techniques, reliable communication is achieved, even in the presence of noise or transmission errors.

SOURCE CODE:

OUTPUT:

CONCLUSION:

_____

_____.

LAB QUESTIONS:

Sr No.	Questions

1	What happens if the character count field is corrupted? How does the receiver detect and handle this error? _____ _____ _____
2	Can you list the advantages and disadvantages of character count framing? _____ _____ _____
3	How does character count framing method compare to other framing techniques in terms of efficiency? _____ _____ _____
4	Why is character stuffing necessary for framing? _____ _____ _____
5	What are the limitations of character stuffing, especially in handling binary data? _____ _____ _____

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PRACTICAL NO. 10

TITLE: Simulate error detection code like Cyclic Redundancy Check (CRC) using C programming

AIM: To simulate and implement CRC for error detection.

OBJECTIVES:

1. Understand error detection and correction.
2. Implement CRC in C and test data integrity.

THEORY:

The **Cyclic Redundancy Check (CRC)** is an error-detecting technique commonly used in digital networks and storage devices to detect accidental changes to raw data. It works by performing a division of the binary data stream using a fixed generator polynomial, resulting in a checksum. This checksum helps detect errors during data transmission or storage.

CRC is typically used for **error detection**, it can be extended or combined with additional mechanisms for basic **error correction**

1) STEPS TO SIMULATE CRC IN C:

1. Data Representation

- Represent the data stream as a binary string or array (e.g., 10101011).
- Use a fixed or user-input generator polynomial, such as 1101.

2. CRC Encoding (Sender Side)

- Append zeros to the data equal to the degree of the generator polynomial (e.g., for 1101, append 3 zeros).
- Perform binary division (using XOR) of the data stream by the generator polynomial to calculate the **remainder (CRC checksum)**.
- Append the remainder to the original data to create the **encoded data**.

3. Data Transmission (Simulating Errors)

- Simulate the transmission process by introducing controlled errors in the encoded data. This can be done by flipping one or more bits in the encoded data.

4. CRC Decoding (Receiver Side)

- Perform the same division operation (XOR) on the received data using the generator polynomial.
- If the remainder is all zeros, the data is valid; otherwise, errors are detected.

5. Error Correction (Optional)

- To implement error correction, you need additional techniques like **Hamming distance** or **error-locating codes**.
- CRC alone is limited in correcting errors, but it can pinpoint the presence of errors for further handling.

<u>2) COMPONENTS OF A CRC SIMULATION IN C:</u>

1. Generator Polynomial

- A binary value used as the divisor in the CRC algorithm.
- Common examples: 1011 ($x^3 + x + 1$), 1001 ($x^3 + 1$).

2. Binary Division

- Division is performed using XOR instead of subtraction.
- Result: Quotient (not used in CRC) and Remainder (CRC checksum).

3. Error Injection

- Simulate errors by flipping bits in the encoded data. Example: Turn 10110101 into 10111101.

4. Validation

- Check the remainder after dividing the received data by the generator polynomial. Non-zero remainder indicates errors.

SOURCE CODE

OUTPUT :

CONCLUSION:

_____

_____.

LAB QUESTIONS:

Sr No.	Questions
1	Explain how the generator polynomial is used in the CRC algorithm. _____ _____ _____ _____ _____ _____

2	What is the difference between error detection and error correction? _____ _____ _____ _____ _____
3	Why is XOR used in the CRC algorithm instead of regular subtraction? _____ _____ _____ _____ _____
4	Why do we append zeros to the data before performing CRC? _____ _____ _____ _____ _____
5	Explain the limitations of CRC in error correction. _____ _____ _____

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